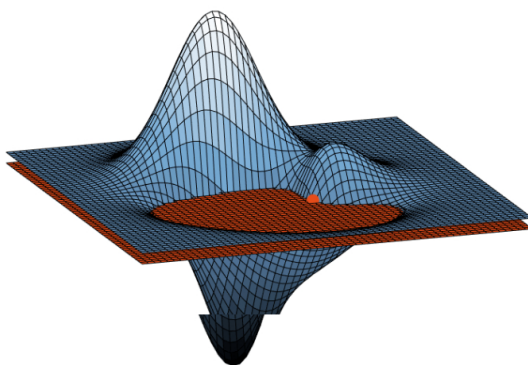


- Now you have seen how to calculate the multivariate Taylor series, and what zeroth, first and second order approximations look like for a function of 2 variables. In this course we won't be considering anything higher than second order for functions of more than one variable.

1 point

In the following questions you will practise recognising these approximations by thinking about how they behave with different x and y , then you will calculate some terms in the multivariate Taylor series yourself.

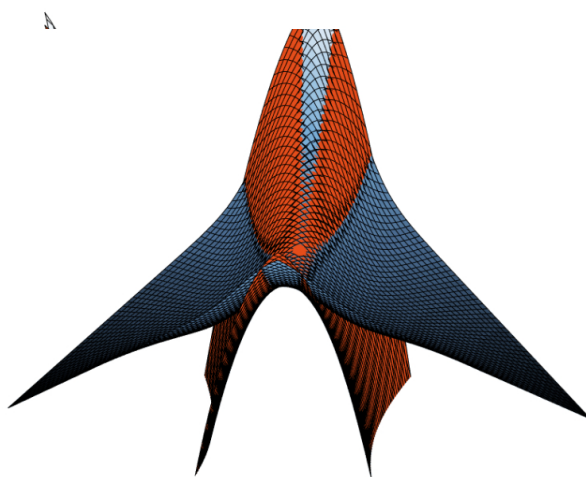
The following plot features a surface and its Taylor series approximation in red, around a point given by a red circle. What order is the Taylor series approximation?



- ☒ Zeroth order
- ☐ First order
- ☐ Second order
- ☐ None of the above

- What order Taylor series approximation, expanded around the red circle, is the red surface in the following plot?

1 point

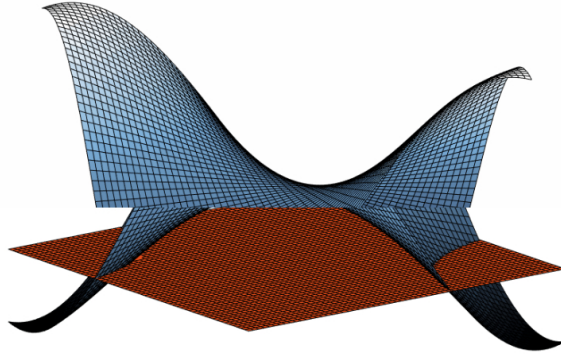


- ☐ Zeroth order
- ☐ First order
- ☒ Second order
- ☐ None of the above

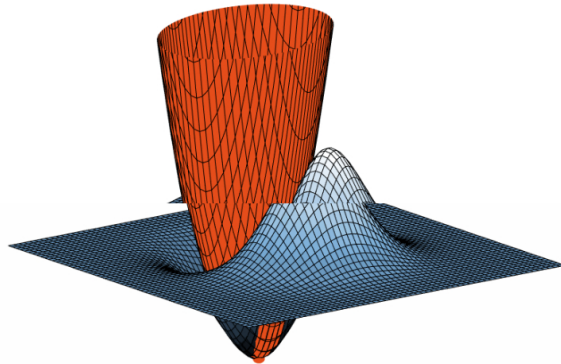
5. Which red surface in the following images is a first order Taylor series approximation of the blue surface? The original functions are given, but you don't need to do any calculations.

1 point

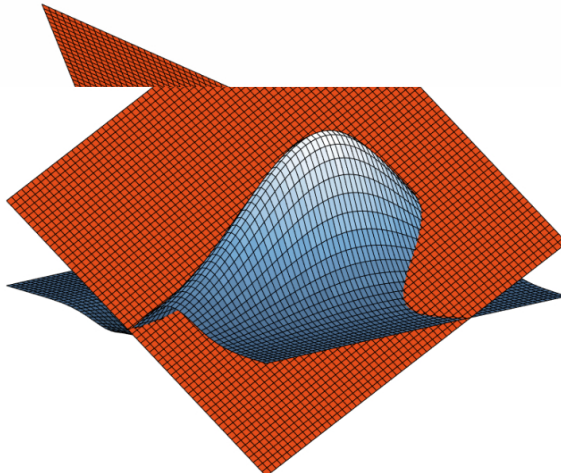
☐ $f(x, y) = \sin(xy/5)$



☐ $f(x, y) = xe^{-x^2-y^2}$

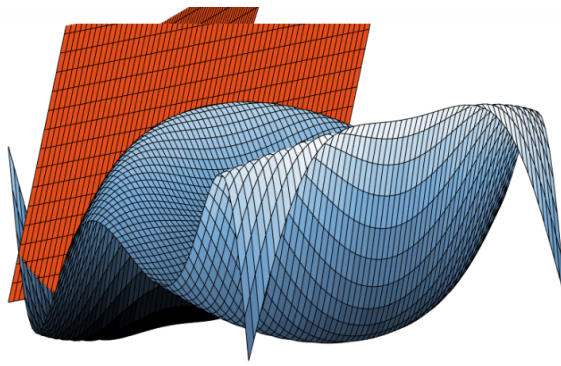


☐ $f(x, y) = (x^2 + 2x)e^{-x^2-y^2/5}$



☒ $f(x, y) = x \sin(x^2/2 + y^2/4)$





4. Recall that up to second order the multivariate Taylor series is given by
 $f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + J_f \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^T H_f \Delta \mathbf{x} + \dots$

1 point

Consider the function of 2 variables, $f(x, y) = xy^2 e^{-x^4 - y^2/2}$. Which of the following is the first order Taylor series expansion of f around the point $(-1, 2)$?

- ☐ $f_1(-1 + \Delta x, 2 + \Delta y) = -4e^{-3} + 16e^{-3}\Delta x - 8e^{-3}\Delta y$
☐ $f_1(-1 + \Delta x, 2 + \Delta y) = 2e^{-33/2} - 63e^{-33/2}\Delta x - 2e^{-33/2}\Delta y$
☐ $f_1(-1 + \Delta x, 2 + \Delta y) = -4e^{-3} - 4e^{-3}\Delta x + 4e^{-3}\Delta y$
☒ $f_1(-1 + \Delta x, 2 + \Delta y) = -4e^{-3} - 12e^{-3}\Delta x + 4e^{-3}\Delta y$

5. Now consider the function $f(x, y) = \sin(\pi x - x^2 y)$. What is the Hessian matrix H_f that is associated with the second order term in the Taylor expansion of f around $(1, \pi)$?

1 point

- ☐ $H_f = \begin{pmatrix} -\pi^2 & \pi \\ \pi & -1 \end{pmatrix}$
☐ $H_f = \begin{pmatrix} -2\pi & -2 \\ -2 & 1 \end{pmatrix}$
☐ $H_f = \begin{pmatrix} -2 & -2\pi \\ 2\pi & 1 \end{pmatrix}$
☒ $H_f = \begin{pmatrix} -2\pi & -2 \\ -2 & 0 \end{pmatrix}$